Irrationality of $\sqrt{2}$

Theorem 1. *The number $\sqrt{2}$ is irrational.*

Proof. We first record the elementary facts that will be used throughout.

Even and odd integers. An integer is called *even* if it has the form $2k$ for some $k \in \mathbb{Z}$, and *odd* if it has the form $2k+1$ for some $k \in \mathbb{Z}$. By the division algorithm with divisor 2, every integer is either even or odd. These two classes are mutually exclusive: if $2a = 2b+1$ for integers a, b, then $2(a-b) = 1$, which is impossible since 2 does not divide 1.

Lowest-terms representation of a rational number. We prove that every rational number can be written as p/q with $q > 0$ and $\gcd(p, q) = 1$.

Let $x \in \mathbb{Q}$, so $x = a/b$ for some $a, b \in \mathbb{Z}$ with $b \neq 0$. Replacing (a, b) by $(-a, -b)$ if necessary, we may assume $b > 0$ without changing the value of x.

Consider the non-empty set

$$S \;=\; \{\, n \in \mathbb{Z} : n \geq 1 \text{ and } n \mid a \text{ and } n \mid b \,\}.$$

Non-emptiness: $1 \in S$ because 1 divides every integer.

Bounded above: Every element $n \in S$ satisfies $n \mid b$ and $b > 0$, so $n \leq b$ (since a positive divisor of a positive integer cannot exceed that integer).

Since S is a non-empty set of positive integers that is bounded above by b, it has a greatest element $d := \max S$; this follows from the fact that every non-empty finite set of integers has a maximum, and S is finite because it is a non-empty set of positive integers bounded above by b.

By definition, $d \mid a$ and $d \mid b$, so we write

$$a = d\,p, \qquad b = d\,q$$

for uniquely determined integers p and q (with $q = b/d > 0$ since $b > 0$ and $d > 0$). Then

$$x \;=\; \frac{a}{b} \;=\; \frac{dp}{dq} \;=\; \frac{p}{q}, \qquad q > 0.$$

It remains to show $\gcd(p, q) = 1$, i.e., that the only positive integer dividing both p and q is 1. Let $c \geq 1$ be a common divisor of p and q, so $p = c\,p'$ and $q = c\,q'$ for some integers p', q'. Then

$$a = dp = d(cp') = (cd)\,p', \qquad b = dq = d(cq') = (cd)\,q',$$

so cd divides both a and b, giving $cd \in S$. Since $d = \max S$, we have $cd \leq d$. Because $c \geq 1$ and $d \geq 1$, this forces $c \leq 1$, hence $c = 1$. Therefore $\gcd(p, q) = 1$.

Main argument. Suppose, for contradiction, that $\sqrt{2}$ is rational. By the lowest-terms property established above, there exist integers p and q with $q > 0$ and $\gcd(p, q) = 1$ such that

$$\sqrt{2} \;=\; \frac{p}{q}.$$

Squaring and multiplying by q^2 gives

$$p^2 \;=\; 2q^2. \tag{1}$$

Hence p^2 is even.

Claim: If n^2 is even then n is even. *Proof of claim:* Suppose n is odd, say $n = 2k + 1$. Then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

which is odd. Since even and odd are mutually exclusive, if n^2 is even then n cannot be odd, so n must be even. $\square$

Applying the claim to p: since p^2 is even, p is even, so $p = 2r$ for some $r \in \mathbb{Z}$. Substituting into (1):

$$(2r)^2 = 2q^2 \implies 4r^2 = 2q^2 \implies q^2 = 2r^2.$$

Hence q^2 is even, so by the same claim, q is even.

Thus $2 \mid p$ and $2 \mid q$, which means 2 is a common divisor of p and q, contradicting $\gcd(p, q) = 1$.

This contradiction shows that $\sqrt{2}$ cannot be rational; therefore $\sqrt{2}$ is irrational. $\square$